SPNet: Spectral Patching End-to-End Classification Network for UAV-Borne Hyperspectral Imagery With High Spatial and Spectral Resolutions

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***Abstract*— In deep learning (DL)-based hyperspectral imagery classification, "spatial patching" is primarily used as a preprocessing for incorporating local spatial information. This operation can help to promote classification accuracy but it is facing new challenges in the unmanned aerial vehicle (UAV)-borne hyperspectral imagery with high spatial and spectral resolutions (H^2 imagery). The ground objects' various spatial scales result in it being challenging to determine the optimal size for the spatial patches. In addition, due to the severe spectral variability and spatial heterogeneity of the H^2 imagery, "spatial patching" only exploits the local spatial information and results in serious salt-and-pepper (SP) noise and isolated areas in the classification maps. In this article, to address these issues, a novel spectral patching network (SPNet) with an end-to-end DL architecture is proposed for UAV-borne H^2 imagery classification. The "spectral patching" approach is proposed to preserve the global spatial information and almost all the spectral information of the original hyperspectral imagery. An end-to-end deep encoder–decoder network is then constructed based on the spectral patching mechanism, which introduces the deep residual network (ResNet) and atrous spatial pyramid pooling (ASPP) modules to extract multiscale high-level semantic information for the H^2 imagery classification. The experimental results obtained with the Wuhan UAV-borne H^2 imagery (WHU-Hi) UAV-borne hyperspectral data set demonstrate that SPNet can achieve state-of-the-art accuracy and visualization performance in the classification of H^2 imagery.**

***Index Terms*— End-to-end, hyperspectral imagery classification, spectral patching, unmanned aerial vehicle (UAV)-borne hyperspectral imagery, Wuhan UAV-borne H^2 imagery (WHU-Hi) data set.**

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I. Introduction

HYPERSPECTRAL images (HSIs) contain tens or even hundreds of continuous narrow bands at a given spectral range, which can accurately characterize the attribute information of ground objects. Compared with space-borne and manned-aircraft-borne hyperspectral imaging systems, a new Earth observation platform—unmanned aerial vehicle (UAV)-borne hyperspectral systems—can simultaneously acquire hyperspectral imagery with high spatial and spectral resolutions (H^2 imagery) [1], [2]. As a result, as shown in Fig. 1(a), the UAV-borne H^2 imagery contains not only fine spectral attribute information of the ground objects but also accurate spatial geometry information. Furthermore, because of the advantages of flexible operation, low cost, and real-time data acquisition, UAV-borne hyperspectral remote sensing systems have been broadly applied in precision agriculture [2]–[4], environmental studies [5], and forestry [3], [6].

Classification is an important aspect of HSI processing and application. However, the research into HSI classification is facing several challenges, such as the complex nonlinear data structure, the curse of dimensionality under the limited training samples, and spectral mixing [7], [8]. Furthermore, the very high spatial resolution of UAV-borne H^2 imagery brings some new challenges for precise H^2 imagery classification. The first challenge is the severe spectral variability and spatial heterogeneity, as shown in Fig. 1(b), with which some classification methods based on spectral information will cause severe salt-and-pepper (SP) noise in the classification results [2]. The other challenge is the big differences in the ground object spatial scales. As a result, in object-oriented classification, it is challenging to choose a suitable segmentation scale. Therefore, spectral–spatial fusion classifiers are needed, which can simultaneously consider the fine spectral attribute information and spatial geometry information.

The traditional spectral–spatial classification methods can lessen the SP noise of classification maps by introducing spatial information [9]. These methods are based on two primary strategies. The first strategy is to cascade the spatial features and spectral information of each pixel as the input of the classifier, where the spatial features include the texture, shape, and object features [10], [11]. The second strategy is to introduce spatial-contextual information to improve the

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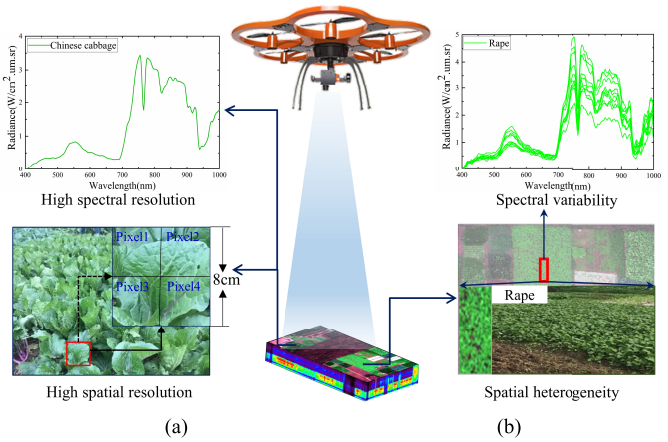

Fig. 1. UAV-borne H^2 imagery with a high spectral resolution (nm level) and a very high spatial resolution (cm level). (a) UAV-borne H^2 imagery (4-cm spatial resolution and 6-nm spectral resolution). (b) Spectral variability and spatial heterogeneity in H^2 imagery.

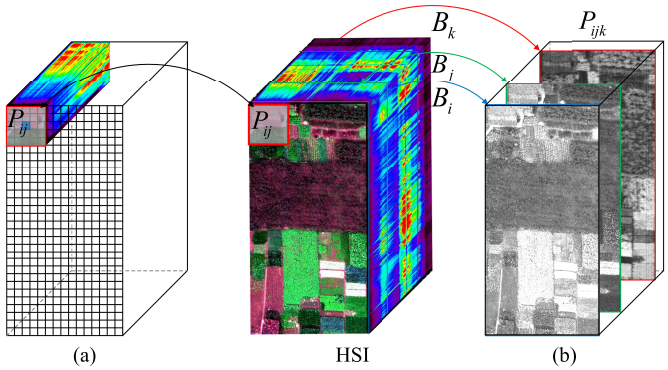

Fig. 2. (a) Spatial patching. (b) Spectral patching. The spectral patch P_{ijk} consists of three bands: B_i, B_j, and B_k.

pixelwise classification accuracy, which can be considered as a postprocessing-based classification method [7]. The typical methods using this strategy are Markov random field (MRF)-based methods [12], [13], conditional random field (CRF)-based methods [14], [15], and extended morphological profiles (EMPs) [16], [17]. Although the effectiveness of the previously mentioned methods in HSI classification has been proved, they are very much reliant on handcrafted features based on the domain knowledge and experience of experts [18]. Furthermore, the traditional classifiers that use handcrafted features as input, e.g., support vector machine (SVM), multiple linear regression (MLR), and random forest (RF), can be regarded as shallow-layer networks, which limits the potential of mining high-level semantic features and the generalization of the classification method for complex scenarios [18], [19].

Recently, to overcome the challenges of handcrafted features and shallow-layer classifiers in the traditional spectral–spatial classification methods, deep learning (DL) has been applied to extract the high-level spectral–spatial features from HSIs automatically. Chen *et al.* [19] first applied the stacked autoencoder (SAE) model to HSI classification, and they designed a network for joint spectral–spatial information extraction and classification. However, the SAE model loses the continuous positional information, to a certain extent, and the network parameters explosively increase with the increase in the fully connected (FC) layers. Convolutional neural networks (CNNs) utilize local connections to learn the images' contextual 2-D spatial information and have achieved improved performances in HSI classification. Makantasis *et al.* [20] used robust principal component analysis (R-PCA) to diminish the spectral dimension of the 3-D patch selected by the neighboring pixel vectors of the labeled pixels as the input of the network. They then used a 2-D CNN for the spatial-dominated information feature extraction and classification. Analogous 2-D CNN networks are Gabor-CNN [21] and PCA-CNN [22]. Furthermore, to fully utilize the continuous spectral and spatial information of ground objects, 3-D CNNs have been introduced to mine the spectral and spatial features from HSIs [22]. In addition, some powerful deep networks have also been applied in HSI classification with spectral–spatial information, including the deep belief network (DBN) [23], the spectral–spatial residual network (SSRN) [24], the 3-D generative adversarial network (3-D GAN) [25], the capsule neural network (CapsNet) [26], the graph convolutional network (GCN) [27], and the cascaded recurrent neural network (CasRNN) [28]. These DL-based methods have all achieved highly competitive performances in HSI classification.

In the above DL models for HSI classification, the primary strategy for utilizing the spatial information is called "spatial patching," which is depicted in Fig. 2(a), where a 3-D patch around the labeled pixel is selected as the input of the DL model for the HSI classification [7]. However, there are two limitations to spatial patching for H^2 imagery classification. First, the optimal size of the spatial patch is challenging to choose for different HSIs and is affected by numerous factors, including the spatial resolution of the imagery and the spatial distribution of the ground objects. The other limitation is that spatial patching only exploits the local spatial information, and the global spatial features of the objects are underutilized. Therefore, holes and isolated areas occur in the HSI classification maps, and this phenomenon is even more serious when classifying high spatial resolution UAV-borne H^2 imagery with a centimeter-level resolution.

In this article, these two limitations of spatial patching are addressed by: 1) the "spectral patching" approach, as depicted in Fig. 2(b), where, for an $M \times N \times L$ HSI, the spectral patching is aimed at generating a set of $M \times N \times 3$ spectral patches as the input of the deep network; compared with spatial patches, each spectral patch preserves all the spatial information of the original HSI, so that the spatial features can be fully exploited during the DL-based classification and 2) the deep residual network (ResNet) and atrous spatial pyramid pooling (ASPP) modules are introduced to extract and fuse the multiscale spatial information, to generate more robust spatial features. The deep ResNet module increases

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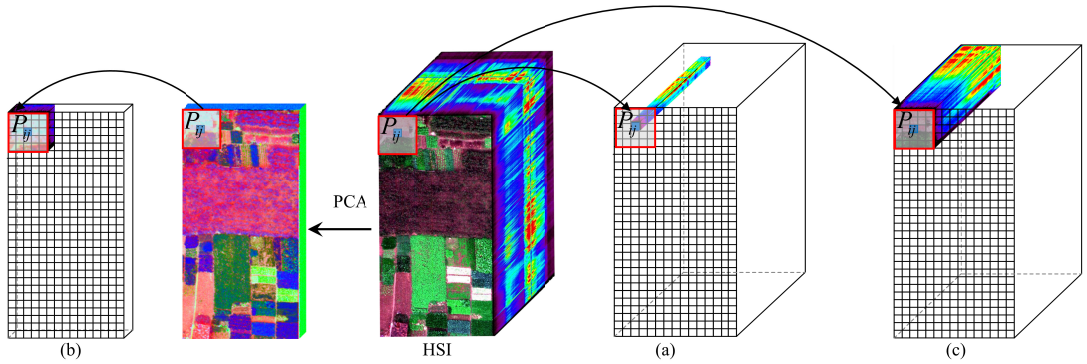

Fig. 3. Three main strategies of spatial patching. (a) Spatial patching based on spectral information. (b) Spatial patching based on local spatial-dominated information. (c) Spatial patching based on local spatial–spectral information.

the depth of the network by employing residual learning, thereby improving the performance of the feature extraction. The ASPP module captures the contextual information of the objects at multiple scales by multiple parallel atrous convolutional filters with different atrous rates [29]. Based on the above strategy, the spectral patching network (SPNet) is proposed for end-to-end UAV-borne H^2 imagery classification, which is inspired by DeepLabv3 plus [30]. SPNet uses three-band spectral patches selected by the spectral patching as the input of the model, and a deep encoder–decoder network is designed to extract the global spatial information. The encoder extracts high-level multiscale contextual information using the ResNet and ASPP modules, and the decoder recovers the boundary details by cascading the low-level feature maps and the high-level semantic information. To offset the spectral information loss of a single spectral patch, the first 15 000 spectral patches with the maximum optimum index factor (OIF) [31], which contain almost all the valuable spectral information from the HSI, are selected for the model training, and the 15 000 predicted probabilistic images from the spectral patches are fused to obtain the final classification result. The main contributions of this article can be summarized as follows.

1) The spectral patching approach is proposed for generating training data for a deep network. Compared with DL for HSI classification methods based on spatial patching, spectral patching fully preserves the global spatial information and almost all the spectral information.
2) A SPNet is proposed for end-to-end UAV-borne H^2 imagery classification. For the challenges of spatial heterogeneity and scale differences in UAV-borne H^2 imagery classification, the deep ResNet and ASPP modules are simultaneously introduced to extract the multiscale high-level semantic information in the encoder. To recover the delicate boundary information of the ground objects, the decoder was designed to cascade the low-level feature maps and high-level semantic information. To minimize the spectral information loss in the spectral patching approach, all the class probability images from the spectral patches are fused to generate the final classification map.
3) The authors have also built the Wuhan UAV-borne H^2 imagery (WHU-Hi) UAV-borne hyperspectral data set[1], which is the first open-source UAV-borne H^2 data set. The WHU-Hi data set is made up of the WHU-Hi-HongHu and WHU-Hi-HanChuan data sets, which were used to verify the effectiveness of the SPNet method. The experimental results confirm that SPNet can achieve state-of-the-art accuracy and visualization performance in the classification of H^2 imagery.

The rest of this article is structured as follows. Section II discusses the related work of DL for HSI classification based on spatial patching. The proposed SPNet method for H^2 imagery classification is presented in Section III. The experimental results and analysis for the WHU-Hi data set are presented in Section IV, followed by a discussion in Section V. Finally, the conclusions and future work are presented in Section VI.

II. Deep Learning for HSI Classification Based on Spatial Patching

At present, there are three main strategies based on spatial patching, which are used to prepare data as the DL model input, as shown in Fig. 3. The three strategies are spatial patching based on spectral information, spatial patching based on local spatial-dominated information, and spatial patching based on local spatial–spectral information.

A. Spatial Patching Based on Spectral Information

Spatial patching based on spectral information is used to select the spectral vectors of the labeled pixels from the HSI as the input of the DL model. For an $M \times N \times L$ HSI, a $1 \times 1 \times L$ spectral vector P_{ij} is selected at position (i, j) and is reshaped into a $1 \times L$ vector to be fed into the DL model. The DL models for HSI classification that use spatial patching based on spectral information include the SAE [19], DBNs [23], 1-D CNNs [22], 1-D GANs [25], [32], and recurrent neural networks (RNNs) [33], [34]. However, because of the spectral variability in HSIs, there will be a

[1]The WHU-Hi data set was collected and processed by the Intelligent Data Extraction and Analysis of Remote Sensing group (RSIDEA group) of Wuhan University, Hubei province, China. It has been published as a benchmark data set on http://rsidea.whu.edu.cn/resource_WHUHi_sharing.htm.

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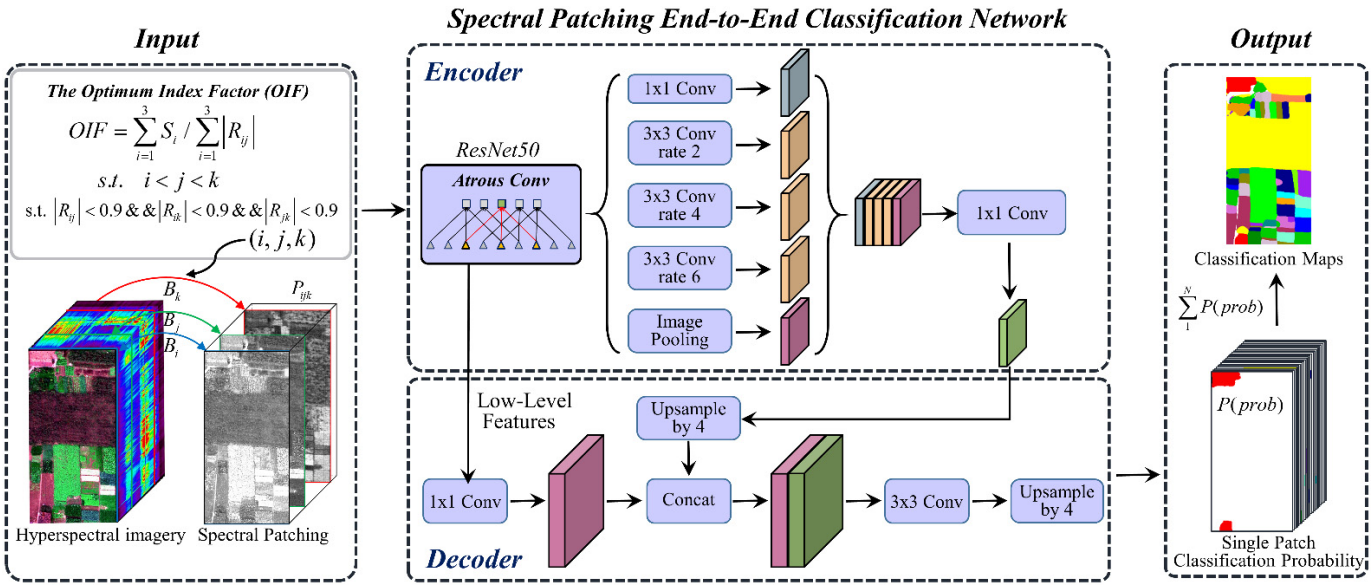

Fig. 4. Flowchart of SPNet for HSI classification.

large amount of SP noise in the classification maps of the DL models that use only the spectral information.

B. Spatial Patching Based on Local Spatial-Dominated Information

Spatial patching based on local spatial-dominated information involves selecting the neighboring pixel vectors of the labeled pixels in the dimensionality reduced HSI as the input of the DL model. The introduction of the spatial information greatly reduces the SP noise in the classification map. For an $M \times N \times L$ HSI, PCA is first implemented for spectral dimension reduction. An $m \times n \times l$ 3-D cube P_{ij} is then selected by the neighboring pixel vectors of the labeled pixels (i, j) in the $M \times N \times l$ image transformed by PCA as the input of the DL model, where l represents the count of the retained principal components. The local spatial-dominated information is often extracted by a 2-D CNN [22], [35], [36], which can extract the discriminative features for HSI classification and reduce the computing requirements. In addition, Chen *et al.* [19] flattened the 3-D cube P_{ij} into a 1-D vector as the input of the SAE model.

C. Spatial Patching Based on Local Spatial–Spectral Information

To adequately utilize the continuous spectral and spatial information of the HSI, spatial patching based on local spatial–spectral information is used to directly select the neighboring pixel vectors of the labeled pixels in the original HSI to be the input of the DL model. For an $M \times N \times L$ HSI, an $m \times n \times L$ 3-D cube P_{ij} is selected at position (i, j) to be fed into the DL model. As a result, the 3-D CNN, which can simultaneously achieve feature extraction in both the spectral and spatial dimensions, is a very effective HSI classification method when combined with spatial patching based on local spectral–spatial information [22]. Moreover, some of the potential DL models that could be used for extracting spectral–spatial features with spatial patching based on local spectral–spatial information are the 3-D GAN [25], SSRN [24], and the fast dense spectral–spatial convolution network (FDSSC) [37].

Researchers have also proposed some more competitive DL models for spectral–spatial feature extraction by combining different spatial patching strategies. Chen *et al.* [19] designed a joint spectral–spatial framework for HSI classification based on the SAE model, where the local spatial-dominated information patch is flattened and then connected with the spectral information patch to be fed into the FC network. The experiments undertaken in this study showed that the classification performance is better than that obtained using a single spatial patching strategy. Xu *et al.* [38] proposed a spectral–spatial unified network (SSUN) for HSI classification, obtaining an excellent classification performance. In this approach, the spectral vector is input into a long short-term memory (LSTM) model to extract the spectral features, and the spatial features are then extracted from the local spatial-dominated information patches. Finally, the spectral–spatial features are fused by an FC layer. Similar dual-branch network DL models have also been proposed in [32], [39], and [40].

III. SPNet for End-to-End HSI Classification

The proposed SPNet is an end-to-end UAV-borne H^2 imagery classification network based on spectral patching, as shown in Fig. 4, which includes data preparation with spectral patching and an end-to-end deep encoder–decoder network. The challenges of DL for HSI classification based on spatial patching are: 1) it is difficult to determine the optimal size of the spatial patches and 2) these DL methods only exploit the local spatial information. Spectral patching is proposed for generating a large number of three-band spectral patches to be the input of the deep network by introducing the OIF, which preserves the global spatial information and

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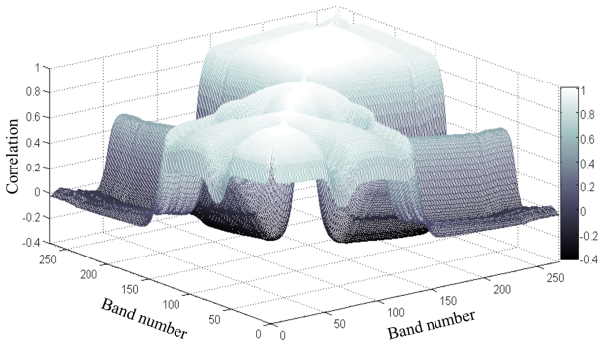

Fig. 5. Correlation matrix of the WHU-Hi-HongHu data set.

almost all the valuable spectral information from the HSI. An end-to-end deep encoder–decoder network is then proposed for the feature extraction and classification. To address the challenges of spatial heterogeneity and scale differences of the ground objects in H^2 imagery, the encoder extracts the high-level multiscale spatial-contextual information from the spectral patches by the deep ResNet and ASPP modules, and the decoder cascades the low-level feature maps and high-level semantic information to recover the ground object boundary information. The softmax classifier is then used to obtain the class probability map for the individual spectral patches. Finally, to adequately utilize the spectral information, all the selected spectral patches with the maximum OIF are used in the model training, and the class probability maps obtained from the spectral patches are fused to obtain the final classification result.

A. Spectral Patching

As shown in Fig. 2, the spectral patching involves selecting three bands from the original $M \times N \times L$ HSI to generate a set of $M \times N \times 3$ spectral patches as inputs of the DL model, where the dimension of the spectral patch channels is inspired by natural imagery. For an $M \times N \times L$ HSI, the spectral patching can generate a spectral patch set $P = \left\{ P_{i,j,k} | i, j, k \in \{1, 2, 3, \ldots, L\} \& i \neq j \neq k \right\}$, containing $A_L^3 = L \times (L-1) \times (L-2)$ spectral patches, where i, j, and k represent the band indices. In general, HSIs have hundreds of bands. For example, the Reflective Optics System Imaging Spectrometer (ROSIS) sensor has 115 bands, the Airborne Visible-Infrared Imaging Spectrometer (AVIRIS) sensor has 224 bands, and the Headwall Nano-Hyperspec sensor used in the experiments in this study has 270 bands, which means that the spectral patch set P can contain millions of spectral patches. However, as shown in Fig. 5, in the correlation coefficient matrix for the UAV-borne HSIs used in this study, the adjacent bands generally have a high correlation. Therefore, there is severe information redundancy in spectral patch set P.

To reduce the redundancy of the spectral patch set, the OIF is introduced to select the subset P_{OIF} with greater information and less redundancy as the input of the network. The OIF is used to select band combinations with a large amount of information and a low correlation, based on the standard deviation and band correlation parameters [31]. Furthermore, the spectral patches selected by OIF are more suitable for human visual interpretation. For a spectral patch $P_{i,j,k}$, the OIF is a statistical value of the ratio between the sum of the standard deviations of the three bands and the sum of the absolute value of the correlation coefficients. However, although the OIF can find the band combination with the largest amount of information in the HSI, the correlation of the band combination will still be high. In the proposed method, the constraints of the band correlation and the values of i, j, and k are used to lessen the relevancy and redundancy in the spectral patches. The formula is given as follows:

$$\begin{aligned} &\mathrm{OIF} = \frac{S_i + S_j + S_k}{\left|R_{ij}\right| + \left|R_{ik}\right| + \left|R_{jk}\right|} \\ &\text{s.t. } i < j < k \\ &\text{s.t. } \left|R_{ij}\right| < 0.9 \&\& \left|R_{ik}\right| < 0.9 \&\& \left|R_{jk}\right| < 0.9 \end{aligned} \tag{1}$$

where S and R represent the standard deviation and correlation matrix value, respectively, and $i, j, k \in \{1, 2, 3, \ldots, L\}$ represent the band indices. Finally, the first 15 000 spectral patches P_{OIF} with the maximum OIF are selected as the input of the deep neural network, as they contain almost all the valuable spectral information from the HSI.

B. End-to-End Deep Encoder–Decoder Network

To adequately utilize the global spatial information, an end-to-end deep encoder–decoder network is introduced in SPNet. As illustrated in Fig. 4, this network contains two primary modules.

1) Encoder: The encoder is aimed at mining high-level semantic features from the three-band spectral patches. However, there are differences in the scales of different class objects, and the scale and spatial texture of the same class in different regions can also be different, which brings challenges to the high-level semantic feature extraction. To overcome the scale difference problem in HSI classification, the deep ResNet module is used as the backbone, combining multiscale spatial information to generate high-level semantic features by the ASPP module. The network architecture of the encoder is shown in Fig. 6.

ResNet solves the vanishing-gradient problem during the deep network training by employing residual learning and bottleneck building blocks and achieves high-level semantic feature extraction with smaller model size and deeper layers [41]. ResNet-50 is used as the backbone of the encoder. The network architecture and the bottleneck building blocks are shown in the top half of Fig. 6. For a three-band spectral patch, the low-level semantic features are first extracted by 7×7 convolution and 3×3 max pooling. The high-level semantic features are then extracted by four blocks, which consists of three, four, six, and three bottlenecks, respectively. To prevent the continuous downsampling from reducing the resolution of the feature map and losing too much detail information, atrous convolution with rate $= 2$ is used in the fourth block of ResNet-50, and the output feature map size is 1/16 of the input data. In addition, as illustrated in Fig. 6 and (2), the bottleneck block applies the shortcut connection

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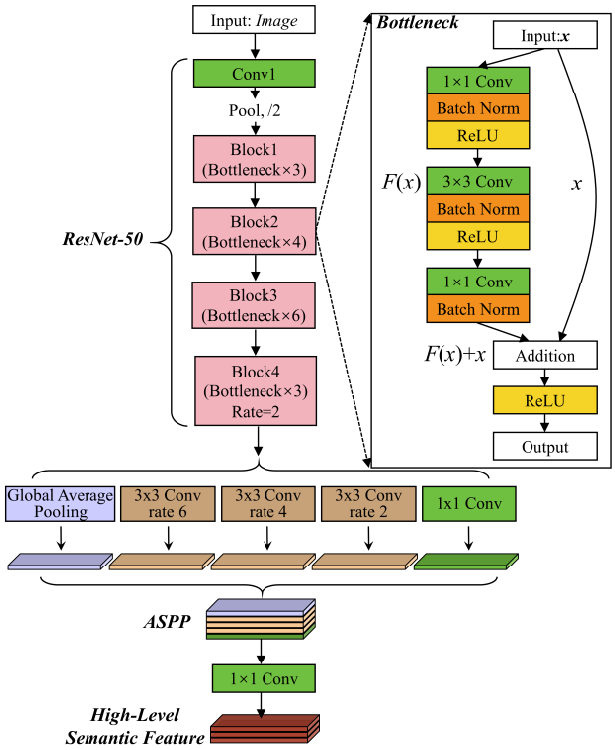

Fig. 6. Encoder architecture.

to add the input feature map x to feature map $F(x)$ as the output y of the bottleneck block

$$y = \sigma(F(x, \{w_i\}) + w_s x) \tag{2}$$

$$F = w_3\sigma(w_2\sigma(w_1 x)) \tag{3}$$

where $F(x, \{w_i\})$ is the residual mapping, w_i represents the ith convolution weight, w_s is used to adjust the channel of input feature map x to the matching feature map $F(x)$, and σ indicates the rectified linear unit (ReLU) activation function.

After ResNet-50, the ASPP module captures the multiscale contextual information of the objects, as well as the image, by applying a 1×1 convolution, and three 3×3 atrous convolutions with rate $= 2, 4, 6$ in separate branches. Global average pooling (GAP) is also introduced to extract image-level features and apply a bilinearly upsampled input size. Finally, the features in the separate branches are then fused to generate a multiscale high-level semantic feature, and 1×1 convolutions are used to reduce the feature dimensionality.

The atrous convolution increases the receptive field size of the convolutional kernel by inserting spaces between the kernel elements, and the number of spaces is constrained by the hyperparameter atrous rate r. For 1-D feature map x, the output $y[i]$ of atrous convolution from $x[i]$ can be defined as

$$y[i] = \sum_{k=1}^{K} x[i + r \cdot k]w[k] \tag{4}$$

where i indicates the pixel location in the y and the corresponding position in x, $w[k]$ indicates the convolutional filter of length K, and r is the atrous rate, which determines the

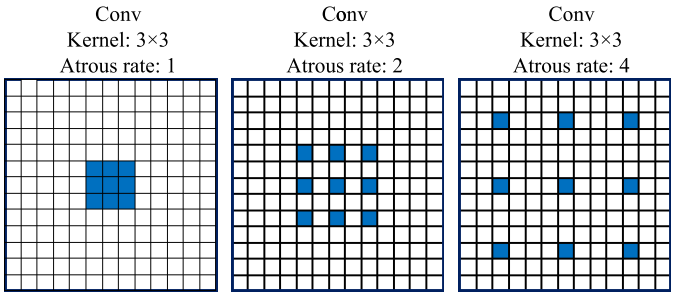

Fig. 7. Atrous convolutions with a kernel size of 3×3 and different rates (atrous rate $= 1, 2, 4$).

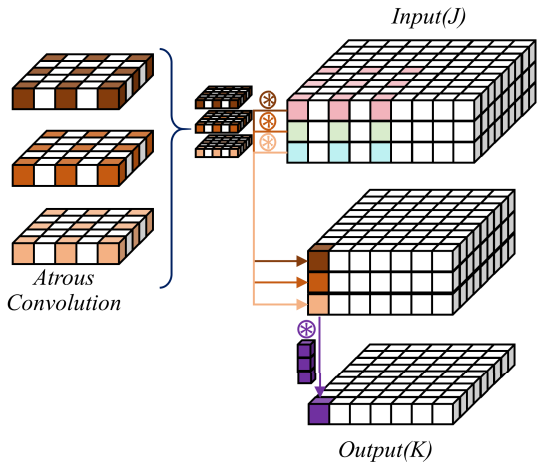

Fig. 8. Atrous depthwise convolutional kernel with a size of 3×3 and atrous rate $= 2$.

sampling stride for the input feature maps. Furthermore, the 2-D atrous convolution can be represented as in Fig. 7.

In order to decrease the complexity of the ASPP module, the atrous depthwise convolution is applied to reduce the model parameters. As shown in Fig. 8, the atrous depthwise convolution applies depthwise separable convolution (DSC) to the atrous convolution, which can separate the mapping of the spatial correlations and channel correlations of the input feature map, to drastically reduce the computational complexity. The DSC first uses the depthwise convolutional filter independently for each input channel, and then, 1×1 pointwise convolution is applied to merge the feature maps from the depthwise convolution. As shown in Fig. 9, the feature map I with L_1 channels is mapped to the feature map J with L_2 channels by a 3×3 convolutional kernel. The number of parameters of the standard convolution is $(3 \times 3 \times L_1) \times L_2 = 9L_1L_2$, while the parameter number of the DSC is $(3 \times 3 \times L_1) + (1 \times 1 \times L_1) \times L_2 = L_1(9 + L_2)$. Therefore, the DSC can effectively diminish the model complexity and computing requirements.

2) Decoder: The decoder is aimed at obtaining the classes of the ground objects while recovering precise boundaries by cascading the low-level feature maps and encoder feature maps. This is because the encoder feature maps contain high-level semantic information to differentiate the different classes, and the low-level feature maps retain the rich spatial

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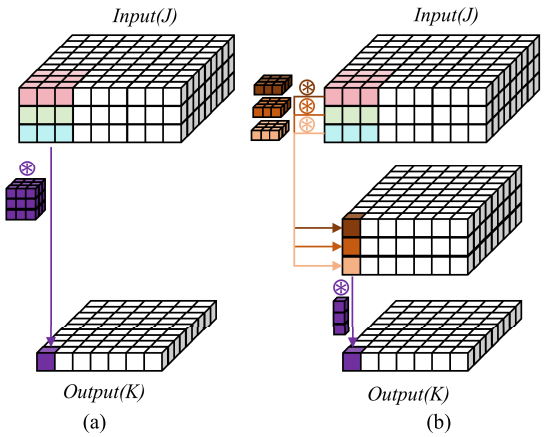

Fig. 9. (a) Standard convolution with a kernel size of 3 × 3. (b) DSC with a kernel size of 3 × 3.

details of the input HSI. First, as shown in Fig. 4, the high-level semantic feature maps from the encoder are 4 × bilinearly upsampled to concatenate with the low-level feature maps from the output of the first block in ResNet-50. Before this, the 1 × 1 convolutional filter is applied to the low-level feature maps to diminish the channel dimensions, the purpose of which is to reduce the proportion of the low-level features in the decoder feature maps and make the model training easier. A few 3 × 3 convolutional filters are then applied to refine the cascaded feature maps, followed by 4 × bilinear upsampling to recover the feature maps to the input image size. Finally, a sparse softmax cross-entropy loss function is employed to solve the loss value, which can reduce memory usage. The loss function of a single spectral patch can be expressed as

$$\text{loss} = -\frac{1}{mn}\sum_{j=1}^{n}\sum_{i=1}^{m}\left(y_i \log(\hat{y}_i) + (1-y_i)\log(1-\hat{y}_i)\right) \quad (5)$$

$$\hat{y}_i = \frac{\exp(z_i)}{\sum_{i=1}^{m}\exp(z_i)} \quad (6)$$

where z represent the output logical value, m and n are the number of land-cover classes and training pixels, respectively, and y_i and $\hat{y}_i$ represent the predicted probability vector and the label of the ith labeled pixel, respectively.

IV. Experiments and Analyses

In this section, the WHU-Hi data set and the experimental settings are first described. The classification performance of the proposed SPNet method with the WHU-Hi data set is then described and analyzed.

A. WHU-Hi Data Set

There are now an increasing number of focused UAV hyperspectral system applications, especially agricultural applications. However, there is still a lack of a public benchmark data set for crop classification based on UAV-borne H^2 imagery. As a result, the authors built an open-source UAV-borne H^2 data set for precise crop classification, which is made up of two individual UAV-borne H^2 data sets, i.e., WHU-Hi-HongHu and WHU-Hi-HanChuan. All the data were acquired using a Headwall Nano-Hyperspec sensor that was equipped on a UAV platform, in farming areas with various crop types in Hubei province, China. The scene of the WHU-Hi-HongHu data set represents a typical fragmented farmland region with various crops planted at a small scale. The scene of the WHU-Hi-HanChuan data set is a rural–urban fringe zone with buildings, water, and cultivated land.

The UAV-borne hyperspectral data preprocessing included radiometric calibration and geometric correction, which were completed in the Hyperspec software provided by the instrument manufacturer. For the radiometric calibration, the raw digital number (DN) value was converted into radiance value by the calibration parameters of the sensor. The geometric correction was then undertaken, based on the collinear equation and the position and attitude information recorded by the global positioning system (GPS)/inertial measurement unit (IMU) module. Ground control points (GCPs) were also used to further promote the geometric correction accuracy of the data.

1) WHU-Hi-HongHu Data Set: The WHU-Hi-HongHu data set was acquired from 16:23 to 17:37 on November 20, 2017, in Honghu City, Hubei province, using a 17-mm focal length Headwall Nano-Hyperspec sensor equipped on a DJI Matrice 600 Pro UAV platform. The experimental area is a typical land fragmentation region, and different cultivars of the same crop are also planted in the region, including Chinese cabbage and cabbage, and *Brassica chinensis* and small *Brassica chinensis*. The UAV flew at an altitude of 100 m, the size of the imagery is 940 × 475, there are 270 bands from 400 to 1000 nm, and the spatial resolution of the UAV-borne hyperspectral imagery is 0.043 m. An overview of this data set is given in Fig. 10(1). The spectra of the 22 classes of ground objects are, respectively, presented in Fig. 10(1.e).

2) WHU-Hi-HanChuan Data Set: The WHU-Hi-HanChuan data set was acquired from 17:57 to 18:46 on June 17, 2016, in Hanchuan, Hubei province, using a 17-mm focal length Headwall Nano-Hyperspec sensor equipped on a Leica Aibot X6 UAV V1 platform. The experimental area contains seven crop species: strawberry, cowpea, soybean, sorghum, water spinach, watermelon, and greens. The UAV flew at an altitude of 250 m, the size of the imagery is 1217 × 303, there are 274 bands from 400 to 1000 nm, and the spatial resolution of the UAV-borne hyperspectral imagery is 0.109 m. Notably, since the WHU-Hi-HanChuan data set was acquired during the afternoon when the solar elevation angle was low, there are many shadow-covered areas in the image. An overview of this data set is given in Fig. 10(2). The spectra of the 16 classes of ground objects are, respectively, presented in Fig. 10(2.e).

B. Experimental Settings

1) Training Sample and Test Sample Initialization: As shown in Tables I and II, for each class, 100 labeled pixels were randomly selected for the model training, and all the remaining pixels were used for the classification accuracy test. Specifically, the proportion of the total number of training pixels was only 0.57% and 0.62% of all the labeled pixels

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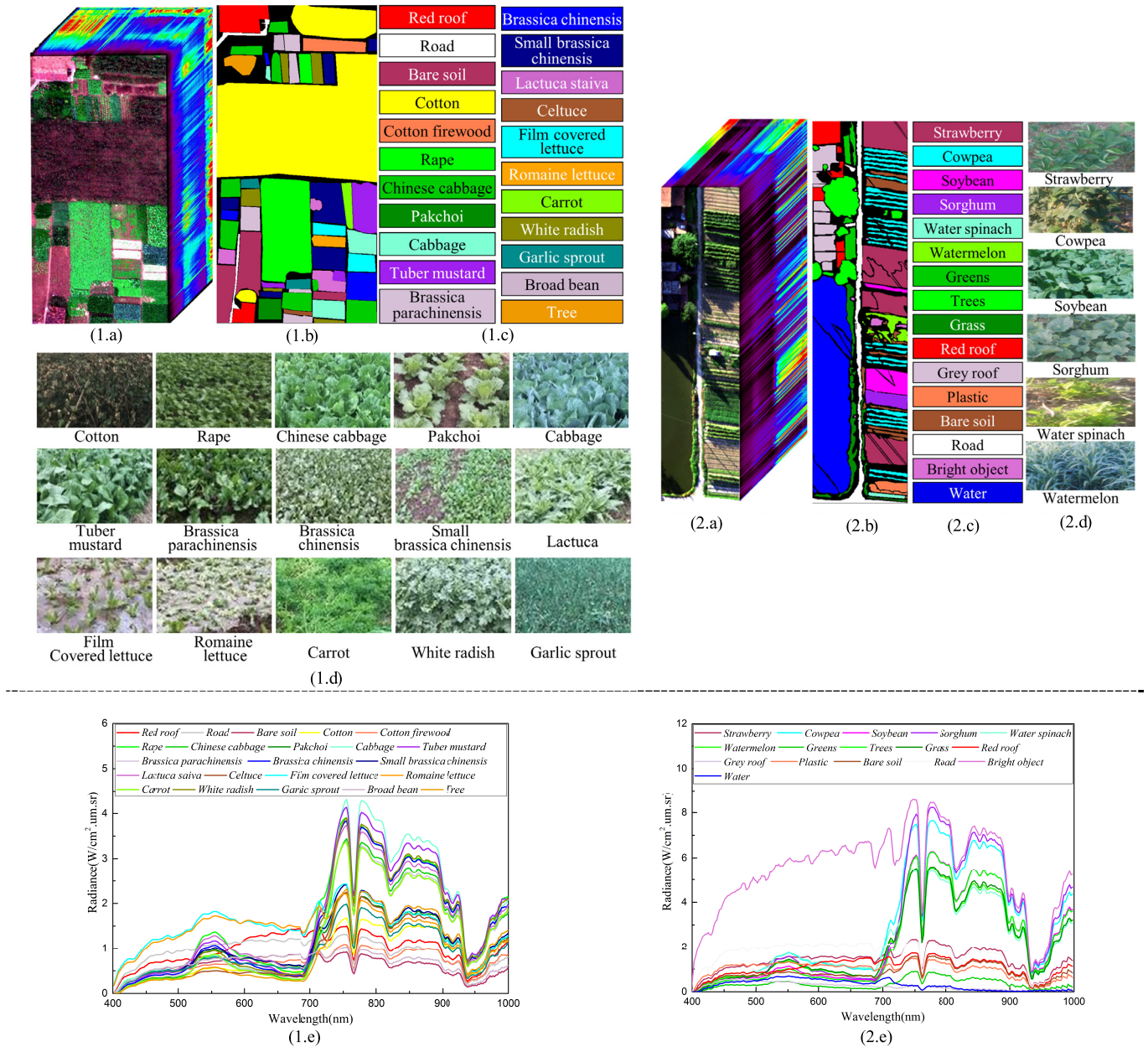

Fig. 10. WHU-Hi data set. (1) WHU-Hi-HongHu data set. (1.a) Image cube. (1.b) G-T data. (1.c) Legend. (1.d) Typical crop photographs in the study area. (1.e) Spectra of the WHU-Hi-HongHu data set land-cover classes. (2) WHU-Hi-HanChuan data set (G-T: ground truth). (2.a) Image cube. (2.b) G-T data. (2.c) Legend. (2.d) Typical crop photographs in the study area. (2.e) Spectra of the WHU-Hi-HanChuan data set land-cover classes.

for the WHU-Hi-HongHu and WHU-Hi-HanChuan data sets, respectively.

2) SPNet Method Parameter Settings: The detailed architecture parameters of SPNet are shown in Table III. The first 15 000 spectral patches with the maximum OIF were selected as the input of SPNet, with the pretrained ResNet-50 as the network backbone. For the SPNet training parameter, we adopted the Nesterov momentum optimizer with momentum $= 0.9$, initial learning rate $= 0.01$, and weight decay $2e-4$. The batch size was 5, and the training epochs are numbered 10.

3) Comparison Algorithms: Five comparison methods were used in the experiments. The first method was spectral-based pixelwise SVM, implemented in the LIBSVM package [42]. The kernel of SVM was the radial basis function (RBF), and the penalty factor C was optimized by fivefold cross-validation. The four other comparison methods were advanced DL models based on spatial patching: SAE with logistic regression (SAE-LR) [19], R-PCA CNN [20], the spectral–spatial attention network (SSAN) [43], and SSRN [24]. SAE-LR is a pioneering DL method in HSI classification, which achieved the best performance in the study of Chen *et al.* [19]. For the PCA images, the first three principal components were retained, and the window size was set as seven. In R-PCA CNN, in order to retain more than 99% of the spectral information, the two experimental data sets were retained with

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TABLE I
CLASS INFORMATION FOR THE WHU-HI-HONGHU DATA SET

No.	Class name	Training samples	Verification samples	No.	Class name	Training samples	Verification samples
C1	Red roof	100	13941	C12	Brassica chinensis	100	8854
C2	Road	100	3412	C13	Small Brassica chinensis	100	22407
C3	Bare soil	100	21721	C14	Lactuca sativa	100	7256
C4	Cotton	100	163185	C15	Celtuce	100	902
C5	Cotton firewood	100	6118	C16	Film covered lettuce	100	7162
C6	Rape	100	44457	C17	Romaine lettuce	100	2910
C7	Chinese cabbage	100	24003	C18	Carrot	100	3117
C8	Pakchoi	100	3954	C19	White radish	100	8612
C9	Cabbage	100	10719	C20	Garlic sprout	100	3386
C10	Tuber mustard	100	12294	C21	Broad bean	100	1228
C11	Brassica parachinensis	100	10915	C22	Tree	100	3940

TABLE II
CLASS INFORMATION FOR THE WHU-HI-HANCHUAN DATA SET

No.	Class name	Training samples	Verification samples	No.	Class name	Training samples	Verification samples
C1	Strawberry	100	44635	C9	Grass	100	9369
C2	Cowpea	100	22653	C10	Red roof	100	10416
C3	Soybean	100	10187	C11	Gray roof	100	16811
C4	Sorghum	100	5253	C12	Plastic	100	3579
C5	Water spinach	100	1100	C13	Bare soil	100	9016
C6	Watermelon	100	4433	C14	Road	100	18460
C7	Greens	100	5803	C15	Bright object	100	1036
C8	Trees	100	17878	C16	Water	100	75301

TABLE III
NETWORK PARAMETERS OF THE SPNET METHOD

Input	The first 15 000 spectral patches with the maximum OIF		
Encoder	*ResNet50*	*Conv*	7×7,64,stride=2
		Max Pooling	3×3, stride=2
		Block1	$\{[1\times1,64]\rightarrow[3\times3,64]\rightarrow[1\times1,256]\}\times3$
		Block2	$\{[1\times1,128]\rightarrow[3\times3,128]\rightarrow[1\times1,512]\}\times4$
		Block3	$\{[1\times1,256]\rightarrow[3\times3,256]\rightarrow[1\times1,1024]\}\times6$
		Block4	$\{[1\times1,512]\rightarrow[3\times3,512]\rightarrow[1\times1,2048]\}\times3$ atrous_rates=2
	ASPP	*Conv*	1×1, 256,stride=1
		Atrous Conv1	3×3, 256, atrous_rates=2
		Atrous Conv2	3×3, 256, atrous_rates=4
		Atrous Conv3	3×3, 256,atrous_rates=6
		GAP	256
	Conv		1×1, 256,stride=1 (Concat feature)
Decoder	*Conv*		1×1, 256,stride=1 (Low-level feature)
	Upsample		×4 (High-level feature)
	Conv		3×3, 256, stride=1 (Concat feature)
	Upsample		×4
Output	Sum all the spectral patch classification probabilities		

the first 30 principal components, and the spatial patch size was set to 5 × 5 × 30. SSAN is a two-branch network based on a spectral–spatial attention mechanism. The spectral branch learns the inner spectral correlations based on the RNN, and the saliency features and spatial relevance are extracted by the CNN in the spatial branch. In this comparison experiment, the input spatial patch size of the spatial branch in SSAN was set to 27 × 27, and four principal components were chosen. SSRN is a superior spectral–spatial HSI classification framework based on 3-D convolution with residual blocks. In this comparison experiment, the size of the spatial patch was set to 7 × 7 × L, where L denotes the band number of the remote sensing image. Finally, to quantitatively evaluate the experimental results, four evaluation indicators are used: the accuracy of each class, the overall accuracy (OA), the average accuracy (AA), and the kappa coefficient (kappa).

C. Experimental Results and Analyses

1) Experiment 1: WHU-Hi-HongHu Data Set: The visualization of classification maps is an important way of judging classification methods. Fig. 11(c)–(h) shows the qualitative classification results of SPNet and the comparison methods (i.e., SVM, SAE-LR R-PCA CNN, SSAN, and SSRN) for the WHU-Hi-HongHu data set. The WHU-Hi-HongHu data set contains 18 kinds of crops, and there is visible spatial heterogeneity for the centimeter-level spatial resolution, which makes it difficult to distinguish the species types using only the spectral information. As shown in Fig. 11(c), because of the similarity of the spectral information among thei different crops, the classification results of SVM show noticeable SP noise, as well as misclassification. For example, the cotton is being seriously misclassified as cotton firewood, and the rape is being seriously misclassified as small *Brassica chinensis*. As displayed in Fig. 11(d)–(h), compared with the shallow pixelwise SVM method using only spectral information, the DL-based methods for HSI classification based on spatial patching introduce local spatial information and automatically extract and fuse in-depth spectral–spatial features. However,

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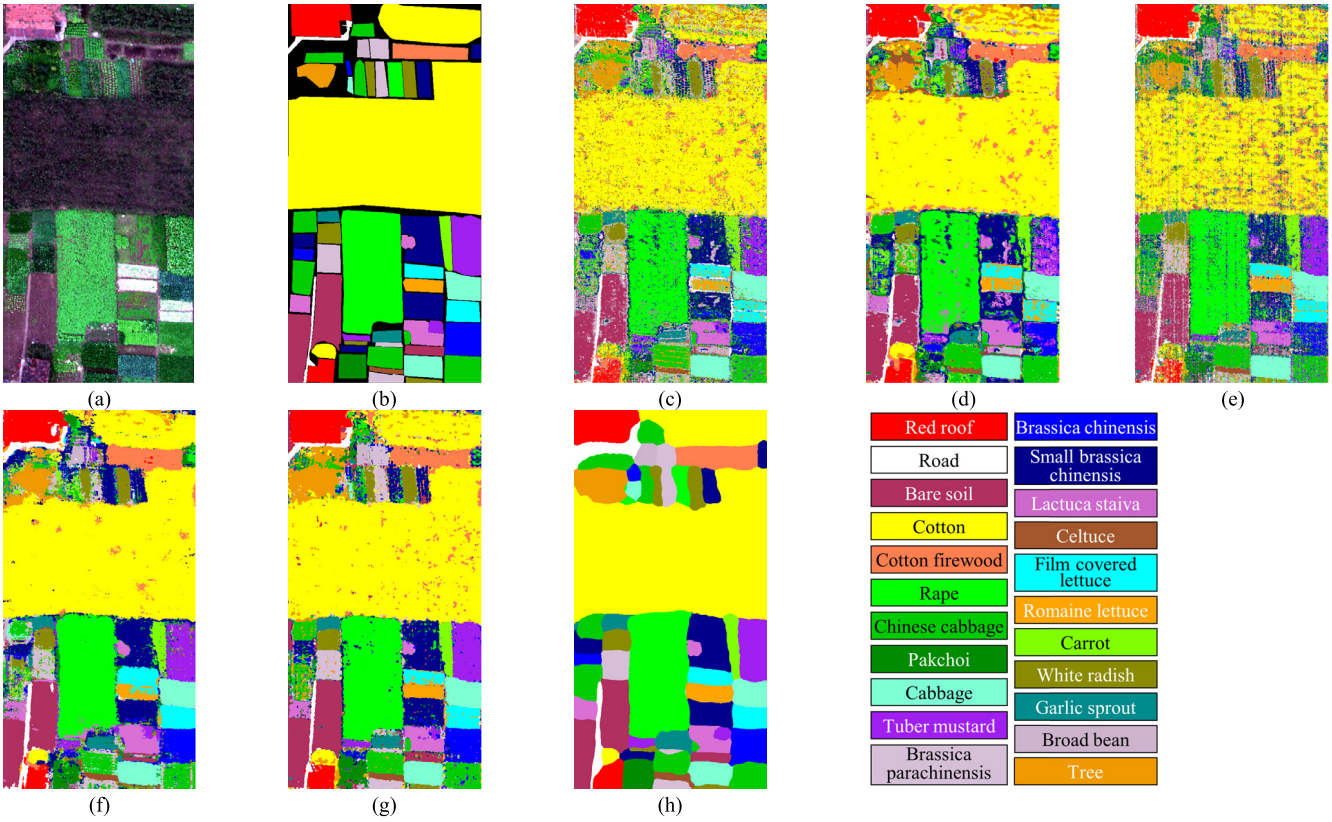

Fig. 11. Classification results for the WHU-Hi-HongHu data set. (a) True-color image (R: Band 108, G: Band 68, and B: Band 27). (b) Ground-truth image. (c) SVM. (d) SAE-LR. (e) R-PCA CNN. (f) SSAN. (g) SSRN. (h) SPNet.

because the patch sizes selected by SAE-LR and R-PCA CNN are relatively small, the utilization of the spatial information is limited for H^2 imagery. Furthermore, the network layers of SAE-LR and R-PCA CNN are shallow, with which it is difficult to extract and fuse in-depth spectral–spatial features. Therefore, there is still very serious SP noise and isolated areas in the visualized classification maps of both the SAE-LR and R-PCA methods, and there is also severe misclassification. The SSAN and SSRN methods adopt deeper networks and advanced feature extraction mechanisms for the HSI classification, which reduces the phenomenon of isolated areas in the visualized classification map. However, due to the severe spectral variability and spatial heterogeneity, there are still some misclassified isolated areas between the cotton/cotton firewood, rape/small *Brassica chinensis*, and Chinese cabbage/*Brassica parachinensis* classes. SPNet makes the best use of the global spatial information with the spectral patching approach, which significantly improves the classification results and reduces the SP noise and isolated areas.

From another perspective, the quantitative accuracy also reflects the performance of the classification methods. As they are obtained by field survey, test samples do not usually include uncertain mixed pixels. In Table IV, the quantitative evaluation results of the different classification methods are listed, where similar results to the visual performance can be seen. The OA of the classification result of SVM is 73.55%, and the OA of the R-PCA CNN classification method is lower than that of SVM due to the small patch size and limited spatial information. The OAs of the SAE-LR, SSAN, SSRN, and SPNet classification methods show improvements of 5.13%, 3.79%, 17.74%, and 23.79%, respectively, over the pixelwise SVM classifier. Furthermore, the SPNet classification method achieves the highest classification accuracy on all the classes of ground objects, achieving a 100% classification accuracy for five classes, including cotton firewood, celtuce, and romaine lettuce. In summary, from these experimental results, it is shown that the SPNet method exhibits a superior classification performance in H^2 imagery classification for the WHU-Hi-HongHu data set.

2) Experiment 2: WHU-Hi-HanChuan Data Set: Fig. 12(c)–(h) shows the classification maps of the different classification methods (SVM, SAE-LR, R-PCA CNN, SSAN, SSRN, and SPNet) for the WHU-Hi-HanChuan data set. The corresponding quantitative performance is provided in Table V. As in Experiment 1, because of the severe spectral variability and spatial heterogeneity in the H^2 imagery, there is severe SP noise in the visualized classification map of SVM. Furthermore, as shown in Fig. 13, the lower reflectivity of the shadow-covered ground objects makes it difficult to achieve classification by SVM, which only uses spectral information. Fig. 14(c) shows that, in the subimage of the SVM classification results, there is serious misclassification in the shadow-covered areas. For example, cowpea is misclassified as greens, and strawberry, road, and water are misclassified as the gray roof classes. Because of the limited training samples, limited spatial information, and shallow network architecture,

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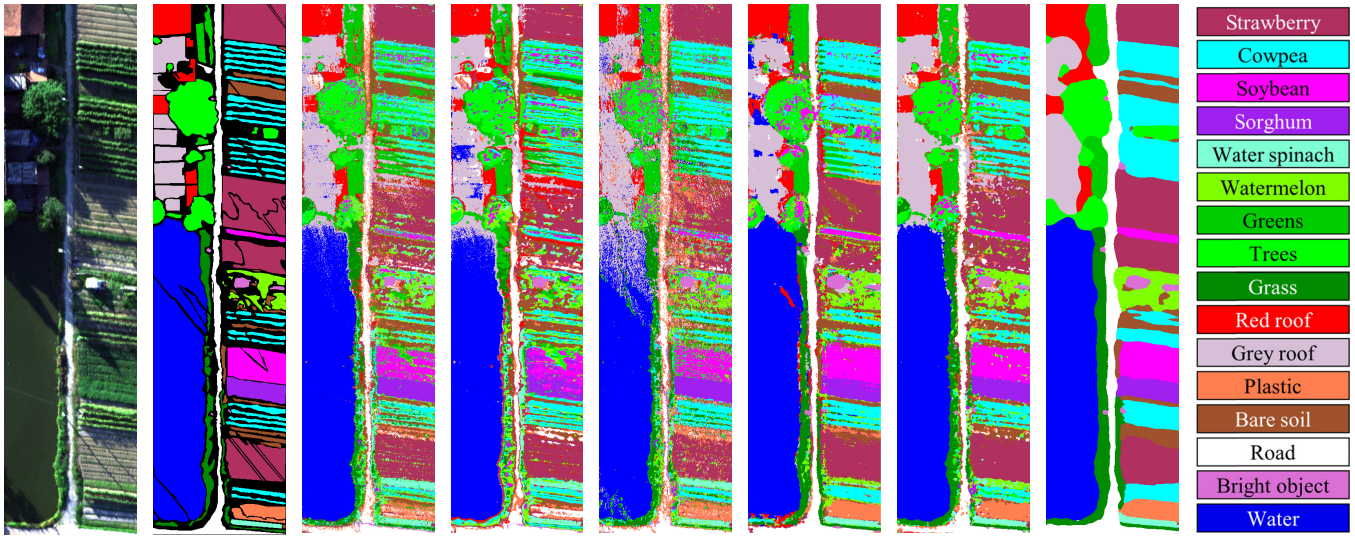

Fig. 12. Classification results for the WHU-Hi-HanChuan data set. (a) True-color image (R: Band 110, G: Band 70, and B: Band 29). (b) Ground-truth image. (c) SVM. (d) SAE-LR. (e) R-PCA CNN. (f) SSAN. (g) SSRN. (h) SPNet.

TABLE IV
CLASSIFICATION ACCURACIES FOR THE WHU-HI-HONGHU DATA SET, WITH THE BEST RESULTS HIGHLIGHTED IN BOLD

	Class	SVM	SAE-LR[19]	R-PCA CNN[20]	SSAN[43]	SSRN[24]	SPNet
Per-class accuracy (%)	Red roof	83.93	92.45	84.33	90.42	97.32	**98.16**
	Road	87.34	83.09	75.23	92.20	94.05	**97.86**
	Bare soil	72.85	79.55	71.37	88.23	95.26	**96.10**
	Cotton	78.96	90.61	67.33	96.10	95.18	**97.76**
	Cotton firewood	77.25	86.09	68.19	96.65	95.08	**100.00**
	Rape	81.95	80.06	82.08	88.18	91.81	**96.73**
	Chinese cabbage	59.25	58.34	64.08	48.44	69.37	**94.84**
	Pakchoi	41.63	40.97	48.94	74.43	86.70	**99.90**
	Cabbage	90.86	90.97	91.54	93.43	97.53	**98.31**
	Tuber mustard	54.08	42.80	62.29	81.57	89.43	**97.45**
	Brassica parachinensis	48.31	13.56	48.86	58.96	90.44	**94.91**
	Brassica chinensis	61.31	51.17	48.02	79.27	83.45	**97.91**
	Small Brassica chinensis	49.86	61.71	39.74	74.51	76.04	**95.87**
	Lactuca sativa	63.78	87.17	64.86	90.90	88.93	**98.00**
	Celtuce	85.92	91.35	91.57	97.56	99.89	**100.00**
	Film covered lettuce	78.01	75.17	83.27	97.68	91.01	**99.47**
	Romaine lettuce	70.65	79.52	90.86	71.48	98.56	**100.00**
	Carrot	79.24	77.99	83.00	87.55	96.34	**99.78**
	White radish	68.22	67.00	74.08	79.56	89.12	**95.89**
	Garlic sprout	77.85	84.11	60.10	93.27	94.89	**100.00**
	Broad bean	74.67	71.99	65.96	92.51	96.82	**99.59**
	Tree	81.14	90.08	77.69	95.51	99.80	**100.00**
	OA (%)	73.55	78.68	68.31	87.34	91.29	**97.34**
	KAPPA	0.6805	0.7365	0.6260	0.8415	0.8910	**0.9665**
	AA (%)	71.23	72.53	70.16	84.93	91.68	**98.11**

the OAs of the SAE-LR and R-PCA CNN classification methods are lower than that of SVM by 5.18% and 2.91%, respectively. Notably, in shadow-covered areas, the SP noise and the isolated areas are more severe in the visualized classification map of R-PCA CNN than that of SVM. The SSAN and SSRN methods show a competitive visual effect, and the classification accuracy is much improved, but, in the classification results for the subimage in Fig. 14, there is still some severe misclassification between the trees/soybean and gray roof/water classes in the shadow-covered areas. However, by taking advantage of the global spatial information, the SP noise and the isolated area phenomenon are greatly relieved in the visualized classification map of the SPNet method, and the highest classification accuracy is achieved in the quantitative evaluation.

V. DISCUSSION

A. Sensitivity in Relation to Training Set Size

The influence of different training and test sets was analyzed using the WHU-Hi-HongHu and WHU-Hi-HanChuan data sets. Different numbers (50, 100, 150, 200, 250, and 300) of labeled pixels were randomly selected from each class for the model training, and the remaining samples were adopted for the model accuracy test. The OAs of the different

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TABLE V

CLASSIFICATION ACCURACIES FOR THE WHU-HI-HANCHUAN DATA SET, WITH THE BEST RESULTS HIGHLIGHTED IN BOLD

Class		SVM	SAE-LR[19]	R-PCA CNN[20]	SSAN[43]	SSRN[24]	SPNet
Per-class accuracy (%)	Strawberry	72.30	68.97	69.53	86.87	85.93	**99.07**
	Cowpea	50.77	65.47	79.94	90.05	78.73	**97.09**
	Soybean	72.66	49.94	72.42	96.20	94.89	**99.51**
	Sorghum	95.68	70.47	97.30	88.84	98.93	**99.26**
	Water spinach	82.91	90.18	82.64	99.73	98.18	**100.00**
	Watermelon	49.09	40.79	52.90	82.27	86.89	**99.26**
	Greens	90.38	79.84	85.23	93.90	91.02	**99.55**
	Trees	61.02	55.68	54.41	60.44	74.34	**98.37**
	Grass	63.19	26.78	67.82	83.10	86.04	**96.22**
	Red roof	89.50	86.75	76.28	84.77	95.06	**98.97**
	Gray roof	93.61	81.82	84.39	83.11	**98.39**	97.85
	Plastic	63.17	34.28	80.72	98.63	96.68	**100.00**
	Bare soil	57.95	37.61	44.09	77.32	79.28	**93.89**
	Road	65.04	71.07	61.93	91.46	83.30	**97.66**
	Bright object	72.49	86.97	90.44	98.07	95.75	**100.00**
	Water	95.56	92.47	84.88	97.20	98.05	**98.33**
OA (%)		77.61	72.43	74.70	88.63	89.82	**98.21**
KAPPA		0.7414	0.6816	0.7099	0.8673	0.8815	**0.9791**
AA (%)		73.46	64.94	74.06	88.25	90.09	**98.44**

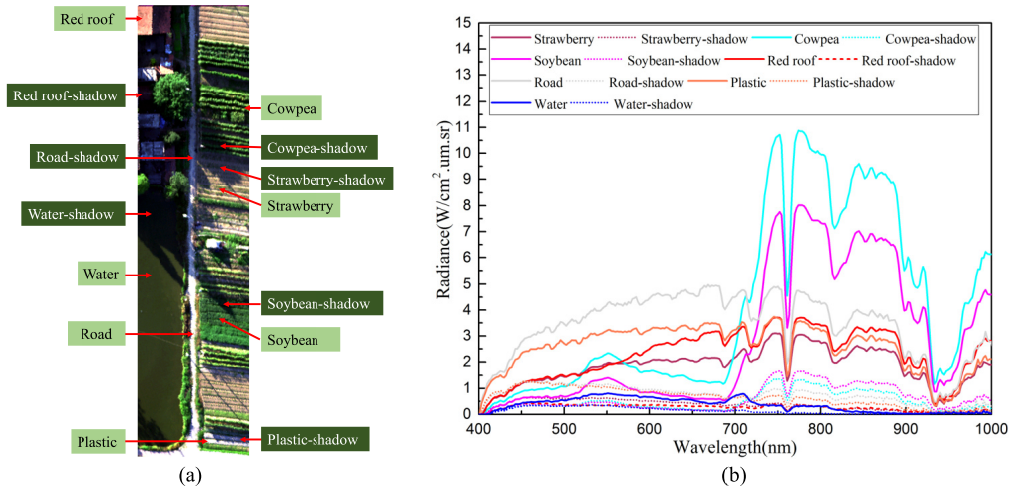

Fig. 13. (a) Shadow-covered ground objects in the WHU-Hi-HanChuan data set. (b) Spectral curves of the ground objects and shadow-covered ground objects.

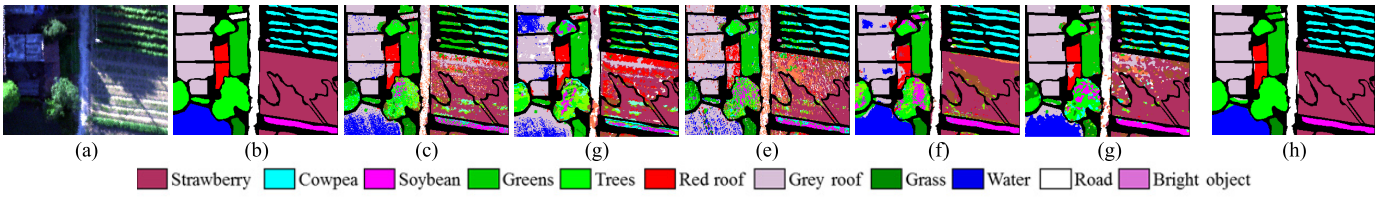

Fig. 14. Classification results for the subimage of the WHU-Hi-HanChuan data set. (a) True-color subimage (R: Band 110, G: Band 70, and B: Band 29). (b) Ground-truth image. (c) SVM. (d) SAE-LR. (e) R-PCA CNN. (f) SSAN. (g) SSRN. (h) SPNet.

classifiers using different numbers of training samples are shown in Fig. 15. It can be seen from the curves that, with the increase in the number of training samples, the performances of the different methods are all improved. SPNet achieves a superior performance under all the training sample sizes, and the OA is better than that of all the comparison methods. Furthermore, SPNet shows a more competitive performance, compared with the comparison methods, when the training sample number for each class is 50.

B. Sensitivity in Relation to the Spectral Patch Number

The influence of different spectral patch numbers for the SPNet method was analyzed using the WHU-Hi-HongHu and WHU-Hi-HanChuan data sets. The different spectral patch numbers (1000, 3000, 5000, 10 000, 15 000, 20 000, and 25 000) were selected as the input data of the SPNet method. The classification accuracies and the band numbers of the HSIs under different spectral patch numbers are shown in Fig. 16. As shown in Fig. 16, it can be found that the OA of SPNet

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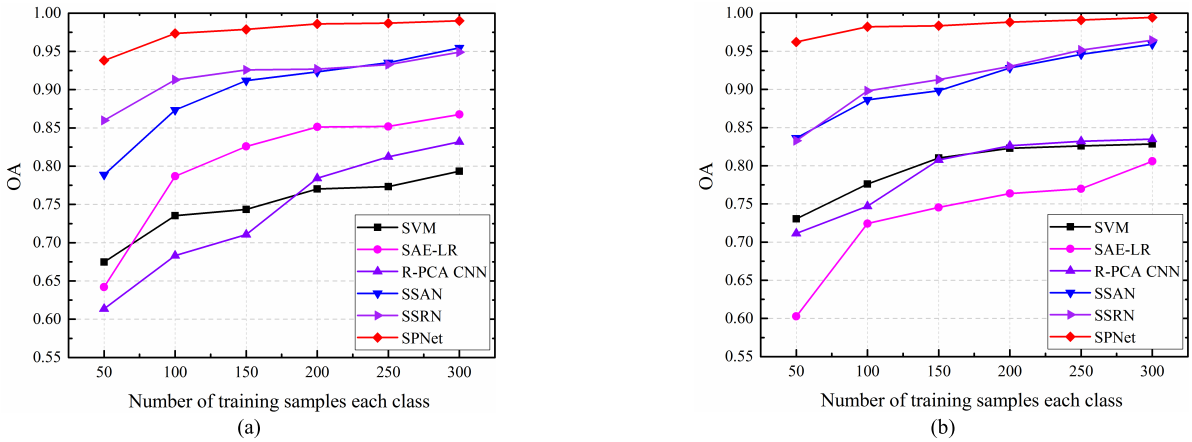

Fig. 15. Effect of different numbers of training samples for SVM, SAE-LR, R-PCA CNN, SSAN, SSRN, and SPNet on (a) WHU-Hi-HongHu data set and (b) WHU-Hi-HanChuan data set.

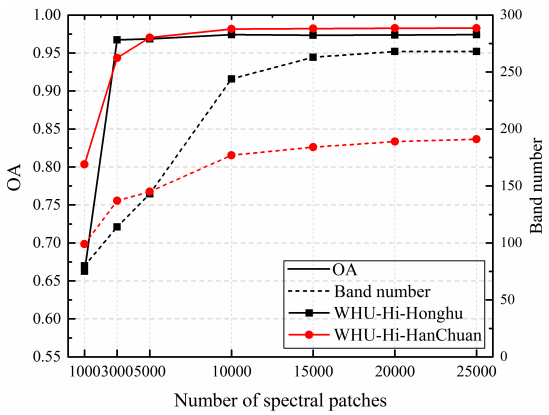

Fig. 16. Effect of different spectral patch numbers for SPNet on the WHU-Hi-HongHu data set and the WHU-Hi-HanChuan data set.

is significantly improved as the number of patches increases from 1000 to 10 000, especially in the range of 1000–3000. When the number of spectral patches surpasses 15 000, the OA of SPNet is completely stable. The OA of SPNet and the band number are highly correlated and show similar curve trends. This is because more bands contain more spectral information, and the OA of SPNet tends to be stable when the spectral information does not significantly increase after the number of spectral patches reaches 15 000. As a result, to balance the required computing resources, the optimal interval of spectral patch number can be determined by analyzing the band number under different numbers of spectral patches.

Furthermore, the band distribution for different spectral patch numbers with the WHU-Hi data set is shown in Fig. 17. For the WHU-Hi-HongHu data set, when the spectral patch number is from 1000 to 5000, the spectral information of the input data is mainly distributed in the near-infrared region. This is due to the WHU-Hi-HongHu data set containing 19 types of plants, and the high reflectivity in the near-infrared region for a plant can better distinguish the difference among plants from a visual perspective. After the number of spectral patches reaches 10 000, the spectral information of the visible spectral region in the input data is significantly increased, especially the highly reflective part of chlorophyll around

TABLE VI

CLASSIFICATION ACCURACIES FOR SPECTRAL PATCHING WITH OTHER DL NETWORKS IN THE WHU-HI DATA SET

Dataset	Deep Learning Network	OA(%)	Kappa	AA (%)
WHU-Hi-HongHu	FCN-VGG32[44]	92.58	0.9146	93.05
	SegNet[45]	94.16	0.9301	94.59
	UNet[46]	95.07	0.9368	95.61
	SPNet	97.34	0.9665	98.11
WHU-Hi-HanChuan	FCN-VGG32[44]	90.60	0.9084	89.98
	SegNet[45]	94.13	0.9514	93.32
	UNet[46]	94.64	0.9560	93.70
	SPNet	98.21	0.9791	98.44

550 nm. When the spectral patch number reaches 15 000, the wavelength range of the band distribution is from 415 to 1000 nm, which covers 97.5% of the wavelength range of the Headwall hyperspectral imaging sensor (400–1000 nm). In addition, there is a low signal-to-noise ratio and little information in the wavelength range of 400 to 415 nm. For the WHU-Hi-HanChuan data set, the wavelength range of the band distribution is from 450 to 903 nm under 15 000 spectral patches, which is due to the wavelength ranges of 400–450 and 900–1000 nm having a low signal-to-noise ratio and serious noise. As a result, the first 15 000 spectral patches with the maximum OIF contain almost all the valuable spectral information.

C. Effectiveness of the Spectral Patching on Other Deep Learning Networks

To investigate the effectiveness of the spectral patching strategy in other DL networks, the classic DL networks were used in this experiment, i.e., FCN-VGG32 [44], SegNet [45], and UNet [46]. In Table VI, the quantitative evaluation results for the spectral patching with other DL networks and the WHU-Hi data set are listed. For the WHU-Hi data set, all the DL networks used in this experiment achieve superior results, which demonstrates that spectral patching is a novel and effective strategy for H^2 image classification. Furthermore, comparing the different network frameworks, SPNet achieves the highest accuracy due to the utilization of the deep ResNet and ASPP modules to eliminate the spatial heterogeneity of H^2 imagery classification.

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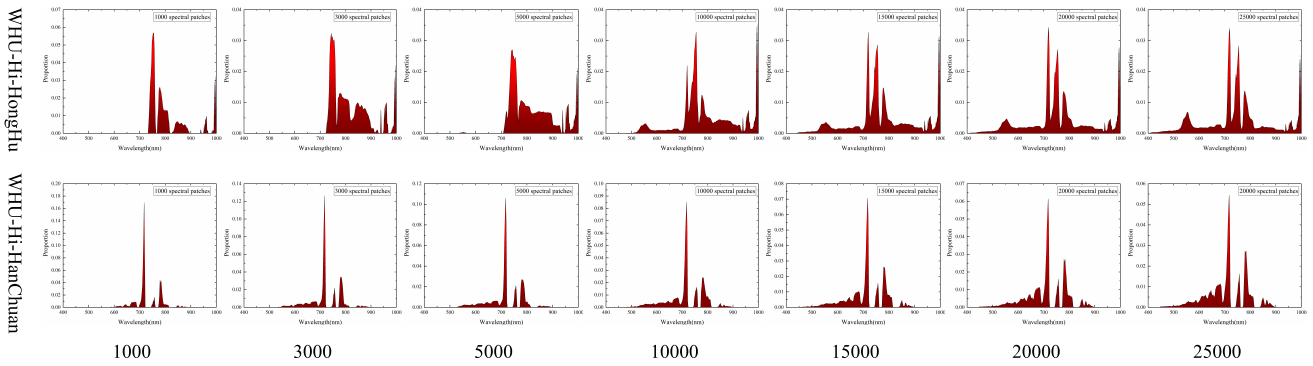

Fig. 17. Band distribution for different spectral patch numbers on the WHU-Hi-HongHu data set and the WHU-Hi-HanChuan data set.

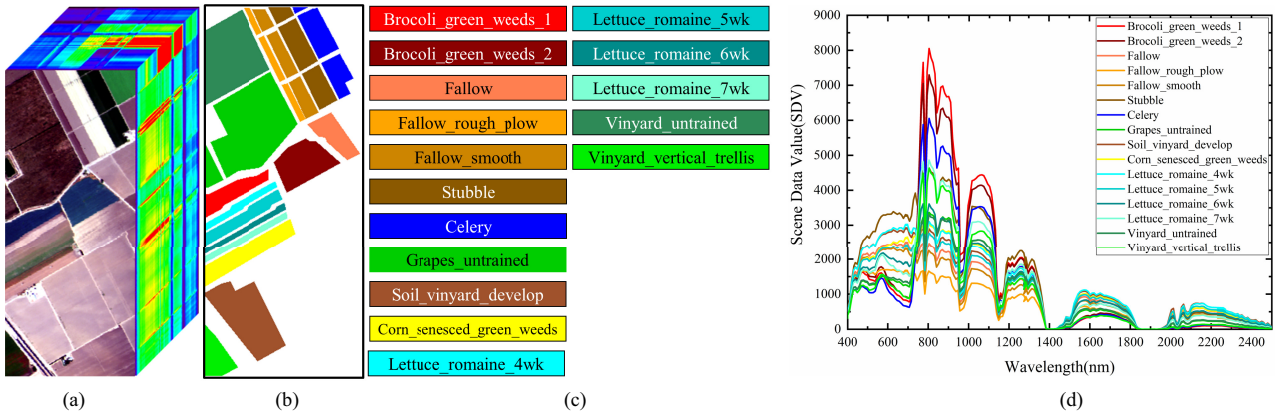

Fig. 18. Salinas data set (G-T: ground truth). (a) Image cube. (b) G-T data. (c) Legend. (d) Spectra of the Salinas data set land-cover classes.

TABLE VII
CLASS INFORMATION FOR THE SALINAS DATA SET

No.	Class name	Training samples	Verification samples	No.	Class name	Training samples	Verification samples
C1	Brocoli_green_weeds_1	50	1959	C9	Soil_vineyard_develop	50	6153
C2	Brocoli_green_weeds_2	50	3676	C10	Corn_senesced_green_weeds	50	3228
C3	Fallow	50	1926	C11	Lettuce_romaine_4wk	50	1018
C4	Fallow_rough_plow	50	1344	C12	Lettuce_romaine_5wk	50	1877
C5	Fallow_smooth	50	2628	C13	Lettuce_romaine_6wk	50	866
C6	Stubble	50	3909	C14	Lettuce_romaine_7wk	50	1020
C7	Celery	50	3529	C15	Vineyard_untrained	50	7218
C8	Grapes_untrained	50	11221	C16	Vineyard_vertical_trellis	50	1757

D. Generality and Applicability of the SPNet Method for Aircraft-Borne Hyperspectral Data Sets

To investigate the generality and applicability of the proposed SPNet method for HSI classification, the Salinas data set was used in this experiment. The Salinas data set was acquired on October 9, 1998, in the south of the city of Greenfield in the Salinas Valley of California, USA, by the AVIRIS sensor equipped on a manned airborne platform. The size of the imagery is 550 × 400, there are 224 bands from 400 to 2500 nm, and the spatial resolution is 3.7 m. An overview of this data set is given in Fig. 18. The spectra of the 16 classes of ground objects are, respectively, presented in Fig. 18(d). In this experiment, 50 labeled pixels for each class were randomly selected as the training samples. The detailed class information for the Salinas data set is provided in Table VII.

Fig. 19(c)–(h) shows the classification maps of the different classification methods (SVM, SAE-LR, R-PCA CNN, SSAN, SSRN, and SPNet) for the Salinas data set. The corresponding quantitative performance is provided in Table VIII. From the perspective of the visual representation, the severe misclassification and isolated area phenomenon are still obvious for the classifiers based on spatial patches, especially for the grapes_untrained and vineyard_untrained classes, due to the spectral and textural features being very similar. The SPNet method eliminates the misclassification and isolated areas in the classification map and achieves the best visual performance. From the perspective of quantitative accuracy, SPNet also achieves the highest classification accuracies (OA, AA, and kappa) in the Salinas data set. As listed in Table VI, the OA of the SPNet method shows improvements of 12.13%, 10.52%, 8.49%, 7.96%, and 6.00% over SVM, SAE-LR, R-PCA CNN, SSAN, and SSRN. In summary, this experiment shows that the proposed SPNet method shows good generality and applicability for the aircraft-borne AVIRIS sensor with a meter-level spatial resolution.

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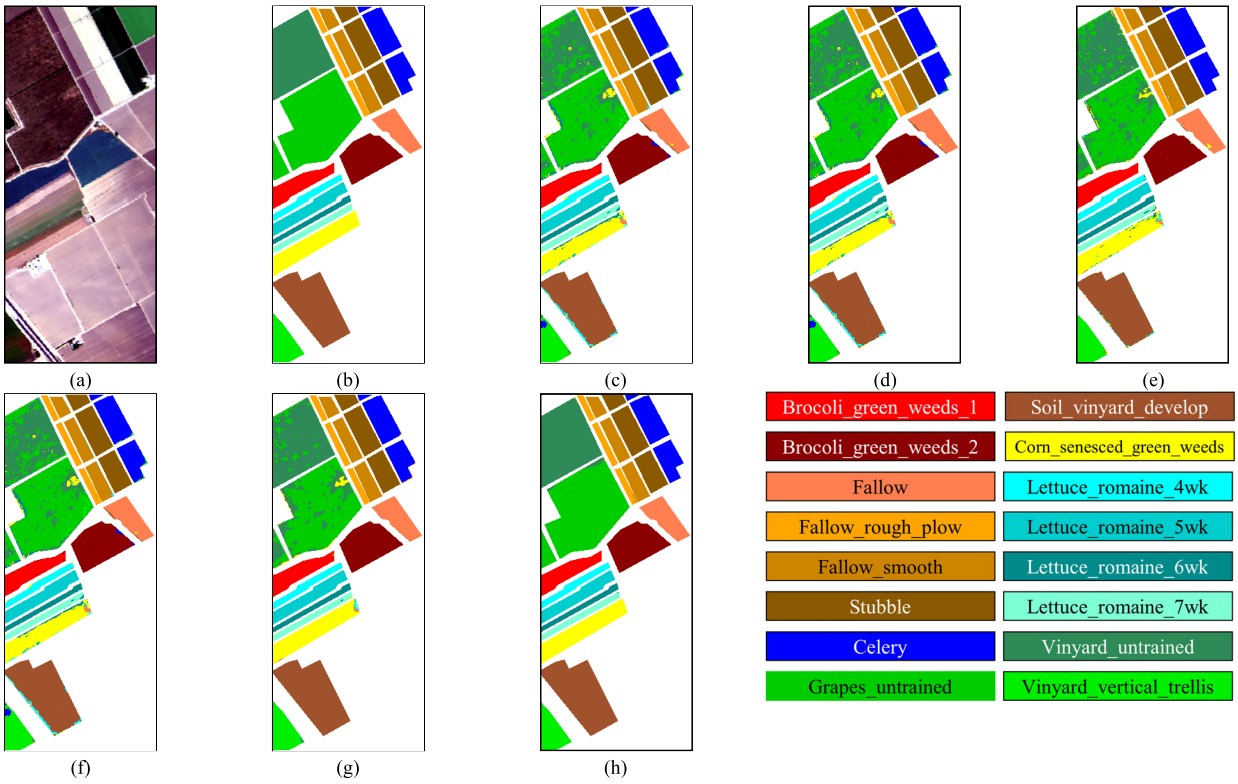

Fig. 19. Classification results for the Salinas data set. (a) True-color image (R: Band 30, G: Band 19, and B: Band 10). (b) Ground-truth image. (c) SVM. (d) SAE-LR. (e) R-PCA CNN. (f) SSAN. (g) SSRN. (h) SPNet.

TABLE VIII

CLASSIFICATION ACCURACIES FOR THE SALINAS DATA SET, WITH THE BEST RESULTS HIGHLIGHTED IN BOLD

	Class	SVM	SAE-LR[19]	R-PCA CNN[20]	SSAN[43]	SSRN[24]	SPNet
Per-class accuracy (%)	Brocoli_green_weeds_1	98.26	96.84	99.90	99.80	99.03	**100.00**
	Brocoli_green_weeds_2	98.94	98.23	99.29	92.90	99.97	**100.00**
	Fallow	93.77	95.69	96.99	99.01	99.74	**100.00**
	Fallow_rough_plow	99.18	99.55	99.70	96.80	**99.85**	97.77
	Fallow_smooth	98.29	96.65	97.60	**100.00**	99.89	99.28
	Stubble	99.54	99.82	99.87	99.85	**100.00**	99.97
	Celery	99.15	97.34	99.52	92.41	99.06	**99.94**
	Grapes_untrained	67.98	80.19	76.37	78.57	77.52	**98.21**
	Soil_vineyard_develop	99.07	94.47	97.97	96.25	99.98	**100.00**
	Corn_senesced_green_weeds	92.81	87.45	90.86	93.31	97.12	**99.91**
	Lettuce_romaine_4wk	93.91	94.70	96.86	95.97	98.92	**100.00**
	Lettuce_romaine_5wk	99.41	99.79	99.47	95.84	**100.00**	99.15
	Lettuce_romaine_6wk	98.38	95.03	99.19	97.92	**100.00**	100.00
	Lettuce_romaine_7wk	94.41	96.86	95.98	99.31	98.43	**99.61**
	Vineyard_untrained	67.48	68.63	80.02	87.46	89.53	**99.89**
	Vineyard_vertical_trellis	95.28	93.06	98.18	98.46	98.69	**100.00**
	OA (%)	87.34	88.95	90.98	91.51	93.47	**99.47**
	KAPPA	85.92	87.71	89.97	90.57	92.74	**99.41**
	AA (%)	93.49	93.39	95.49	95.24	97.36	**99.61**

VI. CONCLUSION

In this article, a new SPNet has been proposed for end-to-end UAV-borne H^2 imagery classification. Instead of spatial patching, spectral patching is proposed for generating training data for the deep network, which fully preserves the global spatial information and almost all the spectral information. With regard to the challenges of the spatial heterogeneity and scale differences for ground objects in UAV-borne H^2 imagery, an end-to-end deep encoder–decoder network is proposed for UAV-borne H^2 imagery classification. The encoder extracts the high-level multiscale spatial-contextual information from the spectral patches by the deep ResNet and ASPP modules, and the decoder cascades the low-level feature maps and the high-level semantic information to recover the ground object boundary information. The softmax classifier is then applied to obtain the class probability maps of the individual spectral patches. Finally, to adequately utilize the spectral information, all the class probability maps obtained by the spectral patches are fused to obtain the final classification map. The experimental results obtained with the WHU-Hi data set confirmed that the proposed SPNet method is a superior classification method for UAV-borne H^2 imagery, and it also has good generality and applicability for aircraft-borne hyperspectral imagery. In the future, a large-scale H^2 benchmark data set will be built for the semantic segmentation and classification of hyperspectral imagery.

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REFERENCES

[1] Y. Zhong *et al.*, "Mini-UAV-borne hyperspectral remote sensing: From observation and processing to applications," *IEEE Geosci. Remote Sens. Mag.*, vol. 6, no. 4, pp. 46–62, Dec. 2018.

[2] Y. Zhong, X. Hu, C. Luo, X. Wang, J. Zhao, and L. Zhang, "WHU-Hi: UAV-borne hyperspectral with high spatial resolution (H^2) benchmark datasets and classifier for precise crop identification based on deep convolutional neural network with CRF," *Remote Sens. Environ.*, vol. 250, Dec. 2020, Art. no. 112012.

[3] T. Adão *et al.*, "Hyperspectral imaging: A review on UAV-based sensors, data processing and applications for agriculture and forestry," *Remote Sens.*, vol. 9, no. 11, p. 1110, Oct. 2017.

[4] P. J. Zarco-Tejada, M. L. Guillén-Climent, R. Hernández-Clemente, A. Catalina, M. R. González, and P. Martín, "Estimating leaf carotenoid content in vineyards using high resolution hyperspectral imagery acquired from an unmanned aerial vehicle (UAV)," *Agricult. Forest Meteorol.*, vols. 171–172, pp. 281–294, Apr. 2013.

[5] L. Wei, C. Huang, Z. Wang, Z. Wang, X. Zhou, and L. Cao, "Monitoring of urban black-odor water based on Nemerow index and gradient boosting decision tree regression using UAV-borne hyperspectral imagery," *Remote Sens.*, vol. 11, no. 20, p. 2402, Oct. 2019.

[6] Y. Tu, M. Bian, Y. Wan, and T. Fei, "Tea cultivar classification and biochemical parameter estimation from hyperspectral imagery obtained by UAV," *PeerJ*, vol. 6, p. e4858, May 2018.

[7] P. Ghamisi *et al.*, "New frontiers in spectral-spatial hyperspectral image classification: The latest advances based on mathematical morphology, Markov random fields, segmentation, sparse representation, and deep learning," *IEEE Geosci. Remote Sens. Mag.*, vol. 6, no. 3, pp. 10–43, Sep. 2018.

[8] P. Ghamisi, J. Plaza, Y. Chen, J. Li, and A. J. Plaza, "Advanced spectral classifiers for hyperspectral images: A review," *IEEE Geosci. Remote Sens. Mag.*, vol. 5, no. 1, pp. 8–32, Mar. 2017.

[9] L. He, J. Li, C. Liu, and S. Li, "Recent advances on spectral–spatial hyperspectral image classification: An overview and new guidelines," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 3, pp. 1579–1597, Mar. 2018.

[10] F. Mirzapour and H. Ghassemian, "Improving hyperspectral image classification by combining spectral, texture, and shape features," *Int. J. Remote Sens.*, vol. 36, no. 4, pp. 1070–1096, Feb. 2015.

[11] L. Zhang, L. Zhang, D. Tao, and X. Huang, "On combining multiple features for hyperspectral remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 3, pp. 879–893, Mar. 2012.

[12] W. Li, S. Prasad, and J. E. Fowler, "Hyperspectral image classification using Gaussian mixture models and Markov random fields," *IEEE Geosci. Remote Sens. Lett.*, vol. 11, no. 1, pp. 153–157, Jan. 2014.

[13] S. Sun, P. Zhong, H. Xiao, and R. Wang, "An MRF model-based active learning framework for the spectral-spatial classification of hyperspectral imagery," *IEEE J. Sel. Topics Signal Process.*, vol. 9, no. 6, pp. 1074–1088, Sep. 2015.

[14] J. Zhao *et al.*, "Spectral-spatial classification of hyperspectral imagery with cooperative game," *ISPRS J. Photogramm. Remote Sens.*, vol. 135, pp. 31–42, Jan. 2018.

[15] P. Zhong and R. Wang, "Jointly learning the hybrid CRF and MLR model for simultaneous denoising and classification of hyperspectral imagery," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 25, no. 7, pp. 1319–1334, Jul. 2014.

[16] P. Ghamisi, J. A. Benediktsson, and J. R. Sveinsson, "Automatic spectral–spatial classification framework based on attribute profiles and supervised feature extraction," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 9, pp. 5771–5782, Sep. 2014.

[17] X. Kang, X. Xiang, S. Li, and J. A. Benediktsson, "PCA-based edge-preserving features for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 12, pp. 7140–7151, Dec. 2017.

[18] S. Li, W. Song, L. Fang, Y. Chen, P. Ghamisi, and J. A. Benediktsson, "Deep learning for hyperspectral image classification: An overview," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6690–6709, Sep. 2019.

[19] Y. Chen, Z. Lin, X. Zhao, G. Wang, and Y. Gu, "Deep learning-based classification of hyperspectral data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2094–2107, Jun. 2014.

[20] K. Makantasis, K. Karantzalos, A. Doulamis, and N. Doulamis, "Deep supervised learning for hyperspectral data classification through convolutional neural networks," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2015, pp. 4959–4962.

[21] Y. Chen, L. Zhu, P. Ghamisi, X. Jia, G. Li, and L. Tang, "Hyperspectral images classification with Gabor filtering and convolutional neural network," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 12, pp. 2355–2359, Dec. 2017.

[22] Y. Chen, H. Jiang, C. Li, X. Jia, and P. Ghamisi, "Deep feature extraction and classification of hyperspectral images based on convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 10, pp. 6232–6251, Oct. 2016.

[23] P. Zhong, Z. Gong, S. Li, and C.-B. Schonlieb, "Learning to diversify deep belief networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 6, pp. 3516–3530, Jun. 2017.

[24] Z. Zhong, J. Li, Z. Luo, and M. Chapman, "Spectral-spatial residual network for hyperspectral image classification: A 3-D deep learning framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 847–858, Feb. 2018.

[25] L. Zhu, Y. Chen, P. Ghamisi, and J. A. Benediktsson, "Generative adversarial networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 9, pp. 5046–5063, Sep. 2018.

[26] F. Deng, S. Pu, X. Chen, Y. Shi, T. Yuan, and S. Pu, "Hyperspectral image classification with capsule network using limited training samples," *Sensors*, vol. 18, no. 9, p. 3153, Sep. 2018.

[27] Z. Gong, P. Zhong, Y. Yu, W. Hu, and S. Li, "A CNN with multiscale convolution and diversified metric for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 6, pp. 3599–3618, Jun. 2019.

[28] R. Hang, Q. Liu, D. Hong, and P. Ghamisi, "Cascaded recurrent neural networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 5384–5394, Aug. 2019.

[29] L.-C. Chen, G. Papandreou, F. Schroff, and H. Adam, "Rethinking atrous convolution for semantic image segmentation," 2017, *arXiv:1706.05587*. [Online]. Available: http://arxiv.org/abs/1706.05587

[30] L.-C. Chen *et al.*, "Encoder-decoder with atrous separable convolution for semantic image segmentation," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Sep. 2018, pp. 801–818.

[31] P. Chavez, G. L. Berlin, and L. B. Sowers, "Statistical method for selecting Landsat MSS," *J. Appl. Photogr. Eng*, vol. 8, no. 1, pp. 23–30, Feb. 1982.

[32] X. Xu, W. Li, Q. Ran, Q. Du, L. Gao, and B. Zhang, "Multisource remote sensing data classification based on convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 937–949, Feb. 2018.

[33] L. Mou, P. Ghamisi, and X. X. Zhu, "Deep recurrent neural networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 7, pp. 3639–3655, Jul. 2017.

[34] H. Wu and S. Prasad, "Convolutional recurrent neural networks for hyperspectral data classification," *Remote Sens.*, vol. 9, no. 3, p. 298, Mar. 2017.

[35] J. M. Haut, M. E. Paoletti, J. Plaza, J. Li, and A. Plaza, "Active learning with convolutional neural networks for hyperspectral image classification using a new Bayesian approach," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 11, pp. 6440–6461, Nov. 2018.

[36] L. Fang, Z. Liu, and W. Song, "Deep hashing neural networks for hyperspectral image feature extraction," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 9, pp. 1412–1416, Sep. 2019.

[37] W. Wang, S. Dou, Z. Jiang, and L. Sun, "A fast dense spectral–spatial convolution network framework for hyperspectral images classification," *Remote Sens.*, vol. 10, no. 7, p. 1068, Jul. 2018.

[38] Y. Xu, L. Zhang, B. Du, and F. Zhang, "Spectral–spatial unified networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 10, pp. 5893–5909, Oct. 2018.

[39] J. Yang, Y.-Q. Zhao, and J. C.-W. Chan, "Learning and transferring deep joint spectral–spatial features for hyperspectral classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 8, pp. 4729–4742, Aug. 2017.

[40] L. Gao *et al.*, "Combining t-distributed stochastic neighbor embedding with convolutional neural networks for hyperspectral image classification," *IEEE Geosci. Remote Sensing Lett.*, vol. 17, no. 8, pp. 1368–1372, Aug. 2019.

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[41] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.
[42] C.-C. Chang and C.-J. Lin, "LIBSVM: A library for support vector machines," *ACM Trans. Intell. Syst. Technol.*, vol. 2, no. 3, pp. 27:1–27:27, Mar. 2011.
[43] X. Mei *et al.*, "Spectral-spatial attention networks for hyperspectral image classification," *Remote Sens.*, vol. 11, no. 8, p. 963, Apr. 2019.
[44] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 3431–3440.
[45] V. Badrinarayanan, A. Kendall, and R. Cipolla, "SegNet: A deep convolutional encoder-decoder architecture for image segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 12, pp. 2481–2495, Dec. 2017.
[46] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. Med. Image Comput. Comput.-Assist. Intervent.*, 2015, pp. 234–241.

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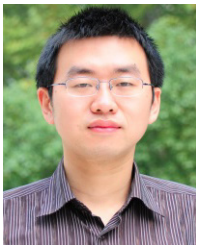

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